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1 Vectors and matrices

1.1 Basic matrix results

Proposition. Matrix multiplication is associative (ie it satisfies $(AB)C = A(BC)$)

Proof. Let A be $n * m$, B be $m * k$, and C be $k * l$.

Then $(AB)_{ij}$ denotes the entry of AB in row i and column j. By matrix multiplication this is $\sum_{r=1}^m A_{ir}B_{rj}$. Also, $((AB)C)_{ij} = \sum_{r=1}^k (AB)_{ir}C_{rj}$. Using the sum for $(AB)_{ir}$ we get $((AB)C)_{ij} = \sum_{r=1}^k \sum_{s=1}^m A_{is}B_{sr}C_{rj}$. If you do the same for $(A(BC))_{ij}$ you will get the same sum. If the elements all agree the matrices are equal.

□

We'll be extra rigorous here and give an obvious proposition.

Proposition. Matrices satisfy $(A+B)C=AC+BC$

Proof. This is kind of obvious since matrix addition just sums the elements, then it is immediate that:

$$\sum_{r=1}^m A_{ir}C_{rj} + \sum_{r=1}^m B_{ir}C_{rj} = \sum_{r=1}^m (A + B)_{ir}C_{rj}$$

Again, the elements agree so the matrices are equal.

□

1.2 Determinants

Theorem. The determinant of a general $n * n$ matrix as given by the formula stated in level 5 and the absolute value of the determinant equals the volume scale factor of a shape transformed by the matrix.

Proof. This has been proven in a video using some visual arguments about vectors. Note that in this proof we show that the sign of the determinant changes when we swap the columns. The fact that whether we have an even or odd number of swaps is well defined (ie for some matrix you can tell which situation we are in) is something which we will prove in Level 7 since the proof is not hard but it is not needed for any of this, and the video basically shows geometrically that it has to be well defined because we know intuitively that the concept of a determinant makes sense. For the three dimensional case you can think of whether the transformed axes are left or right handed (right handed meaning if you take the x-y plane to be parallel to the page with the positive y direction less than 180 degrees counterclockwise from the positive x direction then the z axis will go out of the page).

□

Theorem. The matrix inverse formula we stated in level 5 actually gives the inverse

Proof. This has been proven in the same video as the determinant proof video using the determinant formula and the fact that it is volume.

□

Proposition. A left inverse is also a right inverse, in the sense that if $BA = I$ then $AB = I$.

Proof. Define, for a matrix M , M^{-1} to be such that $M^{-1}M = I$ and we do not yet know that $MM^{-1} = I$. So $B = A^{-1}$.

If $BA = I$ then $\det(A)\det(B) = 1$ so A and B are both invertible. Then $BA = I$ implies $B^{-1}BA = B^{-1}I$ so $A = B^{-1}$

□

1.3 Other results

Proposition. The closest distance between a point (x_1, y_1, z_1) and a plane $Ax + By + Cz + D = 0$ is given by the formula
$$\frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$$

Proof. This is because if a plane is defined by $ax + by + cz + d = 0$ then we know that any point's distance from the plane is given by the length of the line segment normal to the plane, from the plane to the point. We can also rewrite the equation of the plane as $(a, b, c) \cdot (x, y, z) = -d$. We know that the normal has direction (a, b, c) since for any points P, Q on the plane, $(P - Q) \cdot (a, b, c) = -d - (-d) = 0$ and also it is visually obvious that P-Q is perpendicular to the normal of the plane. So the distance from any point R to the plane is the length of the component $R - P$ in the direction of n, where n is normal to the plane and P is a point on the plane. This means it is given by $\frac{|n \cdot (R - P)|}{|n|}$ but $n \cdot P = -d$ so it is given by $\frac{|n \cdot R + d|}{|n|}$, and the result follows.

□

Proposition. Matrices satisfy $(AB)^T = B^T A^T$

Proof. We want to show that $(AB)^T = B^T A^T$ for matrices A and B. This is true because the row i, column j element of AB is given by doing (row i of A) · (column j of B) and so the row j, column i element of $(AB)^T$ is given by that same formula. The row j, column i element of the right hand side of the equation above is equal to (row j of B^T) · (column i of A^T) which by the definition of the transpose is indeed equal to (row i of A) · (column j of B). The matrices are equal element-wise so they are equal.

□

Definition. Vectors $v_1, v_2, \dots, v_n$ are linearly independent if $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$ implies all a_i 's are 0. Essentially, this means that v_n 's point in genuinely new directions and span the full n dimensions.

Definition. We say that $\mathbb{R}^n$ is n -dimensional space.

Definition. We say vectors $v_1, v_2, \dots, v_n$ span $\mathbb{R}^n$ if every vector in $\mathbb{R}^n$ can be written as $a_1v_1 + a_2v_2 + \dots + a_nv_n$ (this is called a linear combination of the v_i 's)

Definition. A basis for $\mathbb{R}^n$ is a linearly independent spanning set. We will prove the “obvious” fact that any basis must have n elements.

Lemma. If $\{v_1, v_2, v_3, \dots, v_n\}$ is a linearly independent set of vectors in $\mathbb{R}^n$ and $\{w_1, w_2, w_3, \dots, w_m\}$ spans $\mathbb{R}^n$, then $n \leq m$

Proof. Since $\{w_1, w_2, w_3, \dots, w_m\}$ spans $\mathbb{R}^n$ write $v_1 = a_1w_1 + a_2w_2 + a_3w_3 + \dots + a_mw_m$. Since $\{v_1, v_2, v_3, \dots, v_n\}$ is linearly independent, $v_1 \neq 0$ so one of the coefficients is non-zero, say the first one. Then this implies we can replace w_1 with v_1 to get a new spanning set.

Now $v_2 = b_1v_1 + b_2w_2 + b_3w_3 + \dots + b_mw_m$, but since the v 's were linearly independent, at least one of the non- v coefficients in this new expression is non-zero, so we can replace this w with v_2 in our spanning set. We can repeat this for every w , and we will never run out since otherwise we will contradict the v 's being linearly independent, meaning $n \leq m$.

□

Proposition. The dimension of $\mathbb{R}^n$ is the same regardless of which basis we use.

Proof. Suppose we have a basis with X vectors and another one with Y vectors for the same vector space, and suppose $X < Y$ (since if $Y < X$ we could apply similar logic without loss of generality): Then the Y basis is linearly independent and the X basis spans the vector space, but $X < Y$ so this contradicts the lemma above.

□

Proposition. a non-invertible matrix cannot correspond to a system of equations with a unique solution

Proof. Let A be non-invertible, then it suffices to show that the equation $Av = 0$ has a non-zero solution, since then given any solution to $Av = w$ we could add a non-trivial solution v of $Av = 0$ to get a second solution.

Since A is not invertible its determinant is 0 which means it collapses everything on to lower dimensions so a basis gets mapped onto a linearly independent set so a non-zero linear combination of these basis elements maps to 0. This is our contradiction.

□

Theorem. The volume of a tetrahedron is $\frac{1}{3}bh$ where b is the area of the base and h is the height. We will prove why this is the case in level 6.1.

Proof. Amazingly, no calculus is needed. Figure 1 shows how you can reconstruct 3 identical pyramids into the cube.

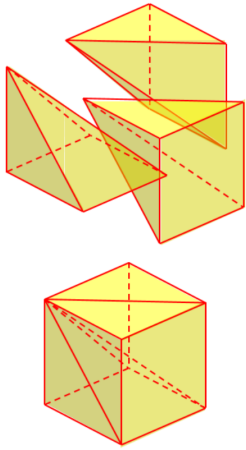


Figure 1

Suppose that the cube has dimensions $1 * 1 * 1$ so that each pyramid in Figure 1 has volume $\frac{1}{3}$. A plane through the apex of a pyramid and a diagonal of the base divides a pyramid into 2 tetrahedra with equal volume which therefore must be $\frac{1}{6}$.

Let a general tetrahedron have two base-edge vectors u and v and a third edge vector w . It is the image of the standard tetrahedron under the linear map with columns u, v, w so its volume is $\frac{|\det[u, v, w]|}{6}$. The corresponding parallelepiped has base area $2b$ and height h so $|\det[u, v, w]| = 2bh$ and so $\frac{|\det[u, v, w]|}{6} = \frac{1}{3}bh$.

We didn't actually use the fact that determinant represents volume here, we could have replaced $\det$ with vol without knowing that it was given by the determinant formula, and the proof would still have worked.

□

1.4 Cross product properties

We want to show that:

1. $A \times B$ is perpendicular to A and B and satisfies the right hand rule (ie $\text{right} \times \text{back} = \text{up}$)
2. $A \times B = -(B \times A)$ (anticommutativity)
3. $A \times (B + C) = A \times B + A \times C$ (distributivity)
4. $|A \times B| = \text{area of parallelogram spanned by } A \text{ and } B$

It will be helpful to think of the cross product as the determinant of the following matrix, since if you actually compute this determinant you can see that it is equal to the formula for the cross product that we assume is true and want to prove these properties from. (Note: i, j, k are the standard unit vectors, and the subscript denotes the coordinate)

$$A \times B = \det \begin{bmatrix} i & A_x & B_x \\ j & A_y & B_y \\ k & A_z & B_z \end{bmatrix}$$

This proves $A \times B = -(B \times A)$ as swapping two columns changes the sign of the determinant, and that $A \times (B + C) = A \times B + A \times C$ since the determinants are linear in columns as shown earlier. Now we can consider the determinant of the following matrix:

$$\begin{bmatrix} A_x & A_x & B_x \\ A_y & A_y & B_y \\ A_z & A_z & B_z \end{bmatrix}$$

Clearly it is zero because it has two columns the same, but another way to see this is to consider expanding it out by the first column, i.e. the determinant is

$$A_x(A_y B_z - B_y A_z) + A_y(B_x A_z - A_x B_z) + A_z(A_x B_y - B_x A_y)$$

But notice, this is actually equal to $A \cdot (A \times B)$, and so we have that $A \cdot (A \times B) = 0$. Applying the same argument to B gives that $A \times B$ is perpendicular to both A and B.

To prove the thing about area, we notice that the sum of the squares of the components of the cross product is the square of the cross product by pythagoras, and that value can also be shown (by a direct computation) to be given by this formula:

$$\det \begin{bmatrix} A_y B_z - B_y A_z & A_x & B_x \\ B_x A_z - A_x B_z & A_y & B_y \\ A_x B_y - B_x A_y & A_z & B_z \end{bmatrix}$$

So now picture this geometrically (See figure 2)

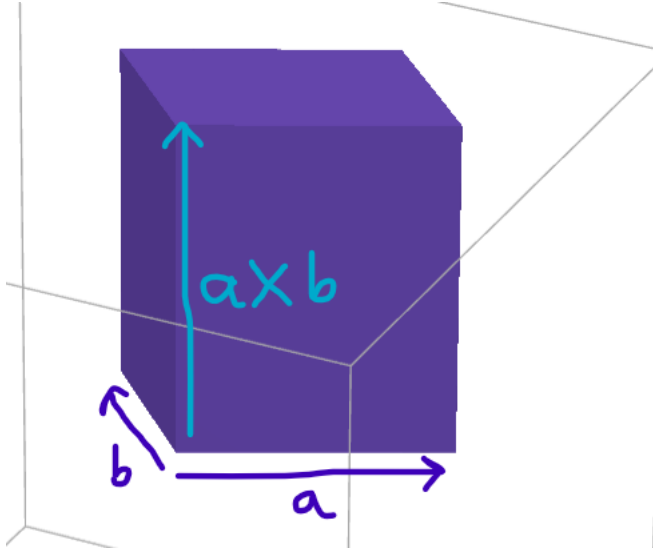


Figure 2

We have a vector $A \times B$ perpendicular to vectors A and B and we want to find the length of $A \times B$. We know that the volume of the parallelepiped spanned by A, B and $A \times B$ (By the volume property of the determinant proven earlier and the determinant of the matrix above) equals the square of the magnitude of the cross product. The determinant (if you expand by the first column) is

$$(A_y B_z - B_y A_z)^2 + (B_x A_z - A_x B_z)^2 + (A_x B_y - B_x A_y)^2$$

By pythagoras, this implies that the determinant of the matrix, which means that since determinant gives the volume of the parallelepiped, the volume of the parallelepiped is $(A \times B)^2$.

If $A \times B = 0$ then property 4 is consistent as the area of the triangle and the vectors magnitude is 0. Otherwise its magnitude is not 0 so we can divide by it which we will do below.

However the volume of this parallelepiped also equals the magnitude of $A \times B$ times the area of the parallelogram spanned by A and B (since $A \times B$ is perpendicular to A and B). Ie, the volume is equal to both $|A \times B|^2$ and $|A \times B|$ (Area of parallelogram spanned by A and B). This means that

$$|A \times B|^2 = |A \times B| (\text{Area of parallelogram spanned by A and B})$$

Cancelling a factor of $A \times B$ on both sides gives the desired result.

1.5 Eigenvalue and Eigenvector properties

Proposition. Let P be an eigenvector matrix of A (the next proposition will show that it is invertible) and D a diagonal matrix with the eigenvalues, then $A = PDP^{-1}$.

Proof. I'll give the intuitive reason it works this way. The reason $A = PDP^{-1}$ is essentially that $(PDP^{-1})v$ takes v as input, and if we take v to be separated into eigenvector components, it first moves v such that those are its normal components (by the P^{-1}), scales everything by the eigenvalue factors (by the D), then moves it back (by the P).

□

Proposition. Eigenvectors with different eigenvalues are actually linearly independent (ie they don't collapse onto a lower dimensional subspace) so the matrix P in the decomposition is actually invertible provided there are n distinct eigenvalues.

Proof. Suppose we have r different eigenvalues and we pick r eigenvectors, one for each eigenvalue. We want to show that they are linearly independent, i.e. none is a linear combination (sum of constant multiples) of the others.

Suppose that they are linearly dependent, meaning there exists constants $a_1, a_2, \dots, a_r$ not all 0 such that $a_1 v_1 + a_2 v_2 + \dots + a_r v_r = 0$ where the v_i 's are our eigenvectors.

Take the p that is the least p such that p of the a 's can be different from 0 with this satisfied.

So now suppose $a_1 v_1 + a_2 v_2 + \dots + a_p v_p = 0$ with all coefficients non-zero. Then

$(A - \lambda_1 I)(a_1 v_1 + a_2 v_2 + \dots + a_p v_p) = 0$ by assumption, but we know that $(A - \lambda_1 I)(a_1 v_1) = 0$ by definition, so $(A - \lambda_1 I)(a_2 v_2 + a_3 v_3 + \dots + a_p v_p) = 0$. Now this means that

$$\begin{aligned} 0 &= (A - \lambda_1 I)(a_2 v_2 + a_3 v_3 + \dots + a_p v_p) \\ &= A(a_2 v_2 + a_3 v_3 + \dots + a_p v_p) - \lambda_1(a_2 v_2 + a_3 v_3 + \dots + a_p v_p) \end{aligned}$$

Therefore, since these are eigenvectors,

$$(a_2(\lambda_1 - \lambda_2)v_2 + a_3(\lambda_1 - \lambda_3)v_3 + \dots + a_p(\lambda_1 - \lambda_p)v_p) = 0$$

This contradicts that p is the least number of non zero coefficients we can have.

□

Theorem. Let S be a symmetric matrix with no repeated eigenvalues, then if we decompose it into its eigenvectors and then normalize them we can write $P^T S P = D$ and $P^T = P^{-1}$.

Proof. It is sufficient to show that the eigenvectors with distinct eigenvalues are perpendicular, as then we see that if the eigenvalue matrix is P then $P^T P = P P^T = I$ by considering taking a dot product between the rows of P^T and the columns of P to multiply the matrices together (since the Eigenvectors are normalized), and we see that if the eigenvectors are normalized and perpendicular this will result in the identity matrix so the transpose of P is the inverse of P .

Note: Specifically, $(P P^T)_{11}$ is essentially dotting the first row of P with the first column of P^T , equivalently dotting the first row of P with itself, which gives 1 since P is orthogonal, and $(P P^T)_{12}$ is essentially dotting the first row of P with the second column of P^T , equivalently dotting the first row of P with its second row, which gives 0 since P is orthogonal. By the same argument with all the entries, $P P^T = I$.

If S is our symmetric matrix and v is an eigenvector with eigenvalue λ then by the definition of eigenvectors we have that $Sv = \lambda v$.

So, suppose we have eigenvalues λ and μ (which are not equal to each other, this is another assumption used in the mathematical statement we are proving) with eigenvectors v and u respectively, then we have

Suppose $Sv = \lambda v$ (1) and $Su = \mu u$ (2). We can multiply both sides of (1) on the left by the row vector u^T and both sides of (2) on the left by the row vector v^T . This will give $u^T Sv = \lambda u^T v$ and $v^T Su = \mu v^T u$, where the constants can be moved out to the front. Now we will take the transpose of both sides of this last equation so we get $(v^T Su)^T = \mu (v^T u)^T$ and then use the relation about matrix transposes from earlier to simplify this to $u^T S^T v = \mu u^T v$. It is here that we use the assumption that S is symmetric, as by definition we have that $S = S^T$ so $u^T Sv = \mu u^T v$.

Since we have $u^T Sv = \lambda u^T v$ from earlier as well, it means $\mu u^T v = \lambda u^T v$ as if they are both equal to $u^T Sv$ they must be equal to each other. Therefore, we have 2 options: Either $\lambda = \mu$ (so the eigenvalues would not be distinct as per our assumption), or $u^T v = 0$. Because of how matrix multiplication works, finding $u^T v$ is actually the same as taking the dot product $u \cdot v$, so we are done and we have proven the perpendicularity as required since dot product is 0 implies perpendicular.

□

Theorem. (Cayley-Hamilton theorem) Every matrix satisfies its own characteristic equation.

Proof. We define $\text{Adj}(M)$ as the transpose of the cofactor matrix (ie, we are in the process of finding the inverse before we divide by the determinant, meaning this always exists as we never run into a division by 0 problem). We have $\det(M) M^{-1} = \text{Adj}(M)$ if M is invertible, and we always have that $\det(M) I = M \text{Adj}(M)$.

We define $B = \text{Adj}(tI - A)$ where A is an $n \times n$ matrix with a characteristic equation in t that is $\det(tI - A) = 0$. Since the elements of an adjugate matrix are built by taking the original matrix and deleting a row and column and finding the determinant (which in the case of $tI - A$ will be polynomials in t with a power of t not exceeding $n - 1$), we know that the elements of B must be polynomials in t of degree up to $n - 1$, so we can write B as

$\sum_{k=0}^{n-1} t^k B_k$ for matrices B_k with coefficients not depending on t .

We know that $\det(tI - A)I = (tI - A)\text{Adj}(tI - A)$ so $\det(tI - A) I = (tI - A) \sum_{k=0}^{n-1} t^k B_k$

On the left hand side, we have $t^n I + c_{n-1}t^{n-1}I + c_{n-2}t^{n-2}I + \dots + c_1tI + c_0I$ where the c 's are the coefficients of the characteristic polynomial of A and on the right hand side we have, by expanding,

$$\begin{aligned} t^n B_{n-1} + t^{n-1}(B_{n-2} - AB_{n-1}) + t^{n-2}(B_{n-3} - AB_{n-2}) + \dots + t(B_0 - AB_1) - AB_0 \\ = t^n B_{n-1} + \sum_{k=1}^{n-1} (t^k(B_{k-1} - AB_k)) - AB_0 \end{aligned}$$

Now we equate the coefficients of powers of t from our two equations. We get that

$$I = B_{n-1}$$

And that

$$c_k I = B_{k-1} - AB_k$$

For k from 1 to $n-1$, and

$$c_0 I = -AB_0$$

Now by multiplying both sides of each equation by A^n , A^k and nothing respectively we get

$$\begin{aligned} A^n &= A^n B_{n-1} \\ c_k A^k &= A^k B_{k-1} - A^{k+1} B_k \\ c_0 I &= -AB_0 \end{aligned}$$

So we have that

$$\begin{aligned} A^n + c_{n-1}A^{n-1} + \dots + c_1 A^1 + c_0 I = \\ A^n B_{n-1} + A^{n-1} B_{n-2} - A^n B_{n-1} + A^{n-2} B_{n-3} - A^{n-1} B_{n-2} + \dots + AB_0 - A^2 B_1 - AB_0 \end{aligned}$$

The latter expression is 0 because clearly every negative term cancels every positive term, and this is equal to the characteristic polynomial of A , where A is an arbitrary matrix. This completes the proof. □

2 Calculus

2.1 Polar coordinate integration

Figure 3 shows the swept area by an arc of a polar curve approximated by thin right angled triangles

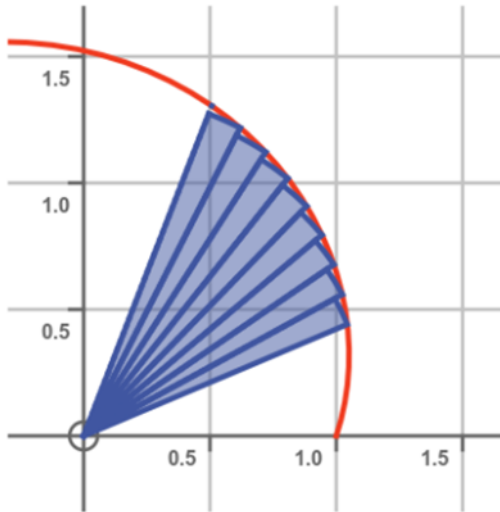


Figure 3

We see from the diagram above that to get a good approximation of the area we are summing the area of right angled triangles which have a base of r , a height of $r \sin(d\theta)$ which is therefore approximately $r d\theta$ with error ratio tending to 1. So the area of these triangles is approximately $\frac{1}{2} r^2 d\theta$. Summing these areas and taking a limit as $d\theta$ goes to 0 is kind of what an integral is, so we get the formula $\int \frac{1}{2} r^2 d\theta$.

2.2 Power series

We will have to do some real analysis preamble to make sense of the results in this section.

Definition. Least upper bound / Supremum essentially means the smallest number greater than or equal to a set of numbers. This always exists if a set is non-empty and bounded above, but it is not necessarily always attained, for example the sequence 0.9, 0.99, 0.999 has a least upper bound of 1 but 1 is not attained in this sequence. It is obvious that any bounded set has a least upper bound, so therefore any increasing sequence which is bounded above converges as it converges to its least upper bound, similarly for decreasing sequences bounded below.

The infimum is the opposite, it is the highest lower bound. It exists for non-empty sets that are bounded below.

If I have a sequence, then the $\liminf$ of the sequence is defined as the limit of the infima of the sequence as I remove more terms at the start. Take some time to think about this definition and the following corollaries if you need to, since I'm not sure how confusing this will be for someone who's never done analysis.

As an example, if my sequence is 0, 2, 0.9, 2, 0.99, 2, 0.999, etc then if I remove terms from the start of this sequence the infimum of this tail sequence will approach 1 so the $\liminf$ is 1. The limit of the sequence does not exist since the terms do not eventually get as close as you want a value. The $\limsup$, which is the opposite of the $\liminf$ (Essentially the limit of suprema) of this sequence is 2. Since the infimum of a sequence does not decrease when you remove terms, the $\liminf$ can be thought of the supremum of the infima of the tails of the sequence (ie the sequences you get after removing terms). In more precise terms, $\liminf_{n \rightarrow \infty} a_n = \sup_n \inf_{k \geq n} a_k$. In fact, the $\liminf$ and $\limsup$ exist exactly whenever the sequence is bounded below or above respectively.

We will now prove some properties of power series.

Proposition. Suppose that $\sum a_n z^n$ converges and $|w| < |z|$, then $\sum a_n w^n$ converges absolutely.

Proof. We know that $|a_n w^n| = |a_n z^n| \left| \frac{w}{z} \right|^n$. Since $\sum a_n z^n$ converges, the terms $|a_n z^n|$ are bounded so we can bound the absolute value by a geometric series which converges, specifically the geometric series $\sum C \left| \frac{w}{z} \right|^n$ for some C which is an upper bound for $|a_n z^n|$.

□

Theorem. All power series, written as $a_0 + a_1z + a_2z^2 + \dots$ with complex coefficients have a unique radius of convergence R with the property that if $|z| < R$ then the series converges absolutely, and if $|z| > R$ then the series diverges. Note that R could be 0 or infinity or anything in between.

Proof. Let $R = \sup\{|z| : a_n z^n \text{ converges}\}$. Then by the lemma, this R has the property we want from the theorem. Specifically, for any $|w| < R$ there is a k such that $a_n w^n$ converges and $|w| < |t| = k < R$ since by definition of supremum we can find something in the $\{|z| : a_n z^n \text{ converges}\}$ set as close as we want to R , then $|w| < k$ and the lemma implies $a_n w^n$ converges absolutely. Furthermore, for any $|w| > R$, $a_n w^n$ must diverge since R is an upper bound for the set of points where it converges. □

Definition. R as defined above is the radius of convergence of a power series.

Theorem. In fact, $R = \frac{1}{\limsup \sqrt[n]{|a_n|}}$

Proof. Suppose $|z| < \frac{1}{\limsup \sqrt[n]{|a_n|}}$. Then $|z| \limsup \sqrt[n]{|a_n|} < 1$. Therefore there exists N and $\varepsilon > 0$ such that

$$\sup_{n \geq N} |z| \sqrt[n]{|a_n|} \leq 1 - \varepsilon$$

by the definition of limsup. Therefore,

$$|a_n z^n| \leq (1 - \varepsilon)^n$$

for every $n \geq N$ which implies by comparison with geometric series that the original power series $\sum a_n z^n$ converges absolutely.

On the other hand, if $|z| \limsup \sqrt[n]{|a_n|} > 1$ then $|a_n z^n| \geq 1$ for infinitely many n so the series diverges. □

Now differentiating (and multiplying by x , which does not affect convergence) changes $|a_n|$ to $|n a_n|$ but the n 'th root of this is $\sqrt[n]{|a_n|} \sqrt[n]{n}$. Note that $\sqrt[n]{n}$ is just $e^{\frac{\ln(n)}{n}}$ which goes to 1 by L'hopital's rule on the exponent. So the derivative of a power series has the same radius of convergence in the general case.

These results will be used when we prove a condition for when series solutions for differential equations are valid.

We will now prove that sums and products of power series converge inside the common radius of convergence of the power series in question, and that compositions of power series converge on a slightly different radius. We write x as if this were not centered at 0 it would just be $x - x_0$.

Proposition. Sums of power series converge inside their common radius of convergence

Proof. We know from earlier that $\sum a_n x^n$ and $\sum b_n x^n$ are absolutely convergent if $|x| < R$. Therefore

$$\sum |a_n + b_n| |x|^n \leq \sum |a_n| |x|^n + \sum |b_n| |x|^n \leq \sum |a_n x^n| + \sum |b_n x^n| < \infty$$

(by the triangle inequality, recall the idea that $|a + b| \leq |a| + |b|$). □

Proposition. Products of power series converge inside their common radius of convergence

Proof. We note that if we have two power series A and B , then the x^n term of AB has contributions from the x^0 term of A with the x^n term of B , the x^1 term of A with the x^{n-1} term of B , the x^2 term of A with the x^{n-2} term of B , and so on. Therefore, the coefficient of the x^n term can be given by $\sum_{k=0}^n a_k b_{n-k}$. So AB can be written as $\sum_{n=0}^{\infty} (\sum_{k=0}^n a_k b_{n-k}) x^n$.

We need absolute convergence to justify rearranging the terms like this, but luckily we have that, since if $|x| < R$ then $(\sum |a_n x^n|)(\sum |b_n x^n|) < \infty$ since each sum converges in the common radius of convergence of the power series a and b.

□

Proposition. Let $g(x)$ be equal to a series of the form $\sum b_n (x - t)^n$ convergent in some interval about t , and let $f(x)$ be a power series convergent in some interval about $g(t)$, then $f(g(x))$ equals a power series convergent in some interval about t .

Proof. Suppose we want to find a power series for $f(g(x))$ where f is $\sum a_n x^n$, g is $\sum b_n x^n$. Set the radius of convergence of f and g to be R_f and R_g respectively. Suppose that $g(x_0) = 0$, because if $g(x_0) = k$ then we can rename f to $f(x - k)$ then g will still be 0 at x_0 .

Let $g^+(x) = \sum |b_n| x^n$. Then we have a closed disc where $|g^+| < R_f$ (possible since $g^+(0) = 0$ and power series are continuous inside their radius of convergence since they are differentiable by level 4 so g^+ can be as small as we want if we make the disc small enough, also making sure the disc does not have a radius larger than R_g). Then inside this disc, the sum $\sum a_n g(x)^n$ is absolutely convergent, since the absolute values are given by $\sum |a_n| |g^+|^n < \sum |a_n| T^n < \infty$ where T is something $< R_f$ so that is where $< \infty$ comes from.

Therefore, by absolute convergence we can safely rearrange or regroup the terms as we like, so composite power series converge inside a finite radius. In the case where f is the exponential function, the power series in question converges in the radius of convergence of g , because g is always within R_f so we just need the other condition that x is within R_g . We will use this fact for the differential equations series solutions proof as well.

□

2.3 Differential equations

Note that integrating factors are unique despite the fact that antiderivatives are not because a constant of integration added to the integrating factor can be moved to a non-zero constant out front of the exponential which can be cancelled on both sides of the equation so it does not matter which antiderivative we use in this context.

We will now discuss how to derive the form of second order differential equations solutions.

Technically this is Second order ordinary linear differential equations with constant coefficients.

To explain why the solutions are of the form they are and why those are the general solutions, I will start by imitating how I think textbooks should introduce it. Currently it feels like you are just following a bunch of rules for the general solution, such as adding x factors for repeated terms, but now I will show where these rules come from. Pretend like you've only seen how to solve first order differential equations using A level techniques, and that you are given the following problem:

Use the substitution $u = \frac{dy}{dx} - 2y$ to solve the differential equation $\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$.

Let's see how we might do this. We observe that $\frac{du}{dx} = \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx}$ by differentiating both sides of the given substitution, which is a natural step as we need to somehow get a $\frac{d^2 y}{dx^2}$ term. Now we can try subtracting $3u$ so that that we have the right $\frac{dy}{dx}$ coefficient of -5, specifically we get $\frac{du}{dx} - 3u = \frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y$. How lucky indeed, we get the original differential equation! To see what is going on, consider this:

$$\begin{aligned} & \frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y \\ &= \frac{d}{dx} \left(\frac{dy}{dx} - 2y \right) - 3 \left(\frac{dy}{dx} - 2y \right) \end{aligned}$$

I'm going to do a notational trick to illustrate the point:

$$= \left(\frac{d}{dx} - 3 \right) \left(\frac{dy}{dx} - 2y \right)$$

Of course, it doesn't make much sense to say " $\frac{d}{dx} - 3$ ", but by "expanding" this to the form before, it hopefully makes sense that what we are effectively doing with the substitution is "factoring" the differential equation, i.e. we use the fact that

$$x^2 - 5x + 6 = (x - 2)(x - 3).$$

Now, let's actually solve it. Recall that we got $\frac{du}{dx} - 3u = \frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y$ so by the original equation we have that $\frac{du}{dx} - 3u = 0$. The general solution to this is $u = Ae^{3x}$. We now reverse the substitution to get $\frac{dy}{dx} - 2y = Ae^{3x}$. We can solve this using the integrating factor method and we will end up getting that $ye^{-2x} = \int Ae^x dx = Ae^x + B$

(Note, in general, if it was not e^x in the integral but something like e^{2x} the A may become a constant multiple of A, but an arbitrary constant multiplied by something is still an arbitrary constant, so we can just write A. It may be a good idea to make a note of this when you use it, but I don't think this is generally necessary for marks on exams.)

Anyway we have that $y = Ae^{3x} + Be^{2x}$. This approach should shed some light on where the form of the solutions to these equations comes from.

Now, we tackle a problem involving repeated roots. We will use the substitution $u = \frac{dy}{dx} + 4y$ to solve $\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + 16y = 0$. We eventually get that $u = Ae^{-4x}$ and therefore $\frac{dy}{dx} + 4y = Ae^{-4x}$. Now the reason that the repeated roots case is different is rooted in what happens when you now try to do the integrating factor method, which gives $\frac{d}{dx}(ye^{4x}) = A$. What happens is that the exponential terms on the right hand side are cancelled. We get that $ye^{4x} = \int A dx = Ax + B$, giving the general solution as $y = (Ax + B)e^{-4x}$. The same thing happens when a term on the right hand side of the differential equation is of the same form as a term in the general solution, but hopefully now it feels less like a rule to follow but something with a clear reason.

Now we look at the case of complex roots. The equation $\frac{d^2y}{dx^2} + y = 0$ gives $y = Ae^{ix} + Be^{-ix}$ as the general *complex* solution for reasons the same as above. We will show that this can be converted into a linear combination of sin and cos, and is in fact equivalent to that, which gives the general *real* solution. We now change our constants to be complex numbers so that

$$\begin{aligned} y &= (A + Bi)e^{ix} + (C + Di)e^{-ix} \text{ where } A, B, C, D \text{ are arbitrary real numbers. This, by Euler's identity, is equivalent to} \\ y &= (A + Bi)(\cos(x) + i\sin(x)) + (C + Di)(\cos(-x) + i\sin(-x)) \\ &= (A + Bi)(\cos(x) + i\sin(x)) + (C + Di)(\cos(x) - i\sin(x)) \\ &= (A + C + Bi + Di)(\cos(x)) + (-B + D + Ai - Ci)(\sin(x)) \end{aligned}$$

And we see that these are still arbitrary complex constants. To get $A + C + Bi + Di = X + Yi$ for arbitrary X and Y and $-B + D + Ai - Ci = U + Vi$ for arbitrary U and V we can set

$$A = \frac{X + V}{2}, B = \frac{Y - U}{2}, C = \frac{X - V}{2}, D = \frac{Y + U}{2}$$

So the general solution is all and only all of the solutions of the form $A\sin(x) + B\cos(x)$ for arbitrary complex numbers A and B. The point is we can convert arbitrary coefficients times conjugate complex exponentials into arbitrary coefficients times sines and cosines, usually without going through the calculation above.

This is now enough theory to understand everything going on with these differential equations.

Proposition. If you find any particular solution to a differential equation, which you can use the ideas above to make an educated guess for without blindly following rules, then that solution plus the complementary function (the solution to the DE with 0 on the right hand side) is the general solution.

Proof. To see this, if we want to find the general solution for y and have a particular solution y_p we can try to find the general solution for $y - y_p$ and then add it to y_p . If the coefficients in the differential equation are constant, or more generally, arbitrary functions of x , so the differential equation is like this

$$A\frac{d^2y}{dx^2} + B\frac{dy}{dx} + Cy = f(x)$$

Where A, B, C can be constants or functions of x , but we know that y_p satisfies the differential equation so

$$A\frac{d^2y_p}{dx^2} + B\frac{dy_p}{dx} + Cy_p = f(x)$$

But now consider this:

$$A \frac{d^2(y - y_p)}{dx^2} + B \frac{d(y - y_p)}{dx} + C(y - y_p) = \left(A \frac{d^2 y}{dx^2} + B \frac{dy}{dx} + Cy \right) - \left(A \frac{d^2 y_p}{dx^2} + B \frac{dy_p}{dx} + Cy_p \right) = f(x) - f(x) = 0$$

Therefore $y - y_p$ is the general complementary function, so we can simply do complementary function plus particular solution.

Also, any “complementary function + particular solution” solution is indeed a solution, by plugging it into the equation and using linearity similar to above.

□

Also, since this is the proof document, I ought to repeat what I said in level 5.1 in the example about coupled differential equations: Note that the result we got (ie what y and z must be if they are a solution) is indeed a solution because the only step that was not completely reversible is differentiating the first equation, but we explicitly found our function z so that our y and z satisfied that first equation.

2.4 Differential equation series solutions

As for series solutions, we don't need to prove anything to be able to find series solutions, so the unproven assertion here is the fact that these solutions are valid on an interval. It would be rather silly if they were not. We use the fact that we can differentiate power series inside the radius of convergence.

We will prove it for the case that differential equations are linear, i.e. either

$$y' = a(x)y + b(x)$$

or

$$y'' = a_1(x)y' + a_0(x)y + b(x)$$

This result will be useful in the level 7 differential equations course.

For the first order case, suppose $y' = a(x)y + b(x)$. This is very easy, as we can find an explicit solution using integrating factors that is therefore given only in terms of integrals of products and compositions of the exponential, a , and b , which all converge within the common radius of convergence of $a(x)$ and $b(x)$ (by the stuff on power series from earlier).

For the second order case, suppose $y'' = a_1(x)y' + a_0(x)y + b(x)$

Note: If the equation is not of this form, then any A level question on it just requires you to find what the power series solution would be if it was valid in some interval, and no information is relevant as to if that is actually the case.

Theorem. A second order differential equation as above, if the coefficients are power series, has a convergent power series solution in at least the common radius of convergence of the power series coefficients.

Proof. We will say $a_1(x) = \sum_{n=0}^{\infty} a_n^{(1)} x^n$, $a_0(x) = \sum_{n=0}^{\infty} a_n^{(0)} x^n$, $b(x) = \sum_{n=0}^{\infty} b_n x^n$. We also assume $y = \sum_{n=0}^{\infty} c_n x^n$, and we will treat the derivative of y as some series which we do not yet know is the true derivative (i.e. $y' := \sum_{n=0}^{\infty} (n+1)c_{n+1} x^n$). After we prove that the power series for y converges when the actual derivative is replaced with this series derivative, we will know that in fact, our power series y satisfies the original differential equation with the true derivative.

We therefore get the following relation by equating the x^n coefficient of both sides of our differential equation.

$$(n+2)(n+1)c_{n+2} = \sum_{p=0}^n (n-p+1)a_p^{(1)}c_{n-p+1} + \sum_{p=0}^n a_p^{(0)}c_{n-p} + b_n \quad (1)$$

Let R be the common radius of convergence of a_1 , a_0 and b . Fix any r with $0 < r < R$. Define the following quantities

$$A_0 := \sum_{p=0}^{\infty} |a_p^{(0)}| r^p$$

$$\begin{aligned}
A_1 &:= \sum_{p=0}^{\infty} \left| a_p^{(1)} \right| r^p \\
B &:= \sup_p |b_p| r^p \\
d_n &:= |c_n| r^n \\
M_n &:= \max_{0 \leq k \leq n} d_k
\end{aligned}$$

Our goal is to prove that M_n is bounded. The reason is let S be the radius of convergence of $\sum c_n x^n$. Then if $r > S$ we have that $\limsup \left(r |c_n|^{\frac{1}{n}} \right) > 1$ and so $|r^n c_n|$ is unbounded.

Multiplying equation (1) by r^{n+2} gives

$$\begin{aligned}
(n+2)(n+1)c_{n+2}r^{n+2} &= \sum_{p=0}^n (n-p+1)a_p^{(1)}c_{n-p+1}r^{n+2} + \sum_{p=0}^n a_p^{(0)}c_{n-p}r^{n+2} + b_n r^{n+2} \\
|c_{n+2}|r^{n+2} &= \frac{\sum_{p=0}^n (n-p+1) \left| a_p^{(1)} \right| |c_{n-p+1}| r^{n+2}}{(n+1)(n+2)} + \frac{\sum_{p=0}^n \left| a_p^{(0)} \right| |c_{n-p}| r^{n+2}}{(n+1)(n+2)} + \frac{|b_n| r^{n+2}}{(n+1)(n+2)} \\
d_{n+2} &\leq \frac{r \sum_{p=0}^n \left| a_p^{(1)} \right| |c_{n-p+1}| r^{n+1}}{n+2} + \frac{r^2 \sum_{p=0}^n \left| a_p^{(0)} \right| |c_{n-p}| r^n}{(n+1)(n+2)} + \frac{r^2 B}{(n+1)(n+2)} \\
d_{n+2} &\leq \frac{r \sum_{p=0}^n \left| a_p^{(1)} \right| r^p |c_{n-p+1}| r^{n-p+1}}{n+2} + \frac{r^2 \sum_{p=0}^n \left| a_p^{(0)} \right| r^p |c_{n-p}| r^{n-p}}{(n+1)(n+2)} + \frac{r^2 B}{(n+1)(n+2)} \\
d_{n+2} &\leq \frac{r A_1}{n+2} M_{n+1} + \frac{r^2 A_0}{(n+1)(n+2)} M_n + \frac{r^2 B}{(n+1)(n+2)}
\end{aligned}$$

By the definitions, $M_{n+2} - M_{n+1} \leq d_{n+2}$ since the amount M_{n+2} can increase by from M_{n+1} is no more than the new term in our list of terms we are finding a maximum from, since they are all positive. Therefore,

$$\begin{aligned}
M_{n+2} - M_{n+1} &\leq \frac{r A_1}{n+2} M_{n+1} + \frac{r^2 A_0}{(n+1)(n+2)} M_{n+1} + \frac{r^2 B}{(n+1)(n+2)} \\
M_{n+2} &\leq M_{n+1} \left(1 + \frac{r A_1}{n+2} + \frac{r^2 A_0}{(n+1)(n+2)} \right) + \frac{r^2 B}{(n+1)(n+2)}
\end{aligned}$$

Let's now assume that M_n is as large as possible, i.e. it is always the case that

$$M_{n+2} = M_{n+1} \left(1 + \frac{r A_1}{n+2} + \frac{r^2 A_0}{(n+1)(n+2)} \right) + \frac{r^2 B}{(n+1)(n+2)}$$

□

Proof. We will say $a_1(x) = \sum_{n=0}^{\infty} a_n^{(1)} x^n$, $a_0(x) = \sum_{n=0}^{\infty} a_n^{(0)} x^n$, $b(x) = \sum_{n=0}^{\infty} b_n x^n$. We also assume $y = \sum_{n=0}^{\infty} c_n x^n$, and we will treat the derivative of y as some series which we do not yet know is the true derivative (i.e. $y' := \sum_{n=0}^{\infty} (n+1)(c_{n+1} x^n)$). After we prove that the power series for y converges when the actual derivative is replaced with this series derivative, we will know that in fact, our power series y satisfies the original differential equation with the true derivative.

We therefore get the following relation by equating the x^n coefficient of both sides of our differential equation.

$$(n+2)(n+1)c_{n+2} = \sum_{p=0}^n (n-p+1)a_p^{(1)}c_{n-p+1} + \sum_{p=0}^n a_p^{(0)}c_{n-p} + b_n$$

Now set $A_r^{(1)} := \sum_{p=0}^{\infty} |a_p^{(1)}| |x|^p$, $A_r^{(0)} := \sum_{p=0}^{\infty} |a_p^{(0)}| |x|^p$, $B_r := \sup_n |B_n| |x|^n$. Note that since $|x| < R$, where R is the common radius of convergence of $a_1(x)$, $a_0(x)$, $b(x)$, these are all finite. We will assume that $|x| < R$ for the rest of this proof. Set $M_n := \max_{0 \leq k \leq n} |c_k| |x|^k$. Using the triangle inequality on the equation above that relates the coefficients, we have that

$$(n+2)(n+1)|c_{n+2}| \leq \sum_{p=0}^n (n-p+1) \left| a_p^{(1)} \right| |c_{n-p+1}| + \sum_{p=0}^n \left| a_p^{(0)} \right| |c_{n-p}| + |b_n|$$

Multiplying both sides by $|x|^{n+2}$ gives

$$\begin{aligned} (n+2)(n+1)|c_{n+2}| |x|^{n+2} &\leq \sum_{p=0}^n (n-p+1) \left| a_p^{(1)} \right| |c_{n-p+1}| |x|^{n+2} + \sum_{p=0}^n \left| a_p^{(0)} \right| |c_{n-p}| |x|^{n+2} + |b_n| |x|^{n+2} \\ (n+2)(n+1)|c_{n+2}| |x|^{n+2} &\leq \sum_{p=0}^n (n-p+1) \left| a_p^{(1)} \right| |c_{n-p+1}| |x|^{n-p+1} |x|^{p+1} + \sum_{p=0}^n \left| a_p^{(0)} \right| |c_{n-p}| |x|^{n-p} |x|^{p+2} + |x|^2 B_r \end{aligned}$$

By the definition of B_r so this inequality still holds.

$$(n+2)(n+1)|c_{n+2}| |x|^{n+2} \leq (n+1) \sum_{p=0}^n \left| a_p^{(1)} \right| |c_{n-p+1}| |x|^{n-p+1} |x|^{p+1} + \sum_{p=0}^n \left| a_p^{(0)} \right| |c_{n-p}| |x|^{n-p} |x|^{p+2} + |x|^2 B_r$$

By the definition of M and A ,

$$(n+2)(n+1)|c_{n+2}| |x|^{n+2} \leq (n+1) M_{n+1} \sum_{p=0}^n \left| a_p^{(1)} \right| |x|^{p+1} + M_n \sum_{p=0}^n \left| a_p^{(0)} \right| |x|^{p+2} + |x|^2 B_r$$

$$(n+2)(n+1)|c_{n+2}| |x|^{n+2} \leq (n+1) M_{n+1} A_r^{(1)} |x| + M_n A_r^{(0)} |x|^2 + |x|^2 B_r$$

$$|c_{n+2}| |x|^{n+2} \leq \frac{M_{n+1} A_r^{(1)} |x|}{n+2} + \frac{M_n A_r^{(0)} |x|^2}{(n+2)(n+1)} + \frac{|x|^2 B_r}{(n+2)(n+1)}$$

But by the definitions, $M_{n+2} - M_{n+1} \leq |c_{n+2}| |x|^{n+2}$ since the amount M_{n+2} can increase by from M_{n+1} is no more than the new term in our list of terms we are finding a maximum from, since they are all positive. Therefore,

$$M_{n+2} \leq M_{n+1} + \frac{M_{n+1} A_r^{(1)}}{n+2} + \frac{M_n A_r^{(0)}}{(n+2)(n+1)} + \frac{|x|^2 B_r}{(n+2)(n+1)}$$

Now set $S_n := M_n + C$ where $C := \frac{|x|^2 B_r}{A_r^{(1)} + A_r^{(0)}}$. In the problem case that $A_r^{(1)}$ and $A_r^{(0)}$ are both 0, our differential equation is $y'' = b$ so the thing we are trying to prove follows from basic properties of power series.

We get

$$\begin{aligned} S_{n+2} - C &\leq S_{n+1} - C + \frac{S_{n+1} A_r^{(1)}}{n+2} - \frac{C A_r^{(1)}}{n+2} + \frac{S_n A_r^{(0)}}{(n+2)(n+1)} - \frac{C A_r^{(0)}}{(n+2)(n+1)} + \frac{|x|^2 B_r}{(n+2)(n+1)} \\ S_{n+2} &\leq S_{n+1} + \frac{S_{n+1} A_r^{(1)}}{n+2} - \frac{C A_r^{(1)}}{(n+2)(n+1)} + \frac{S_n A_r^{(0)}}{(n+2)(n+1)} - \frac{C A_r^{(0)}}{(n+2)(n+1)} + \frac{|x|^2 B_r}{(n+2)(n+1)} \end{aligned}$$

Therefore, simplifying, we have

$$S_{n+2} \leq S_{n+1} \left(1 + \frac{A_r^{(1)}}{n+2} \right) + \frac{S_n A_r^{(0)}}{(n+2)(n+1)}$$

Since the three terms on the right of the previous equation cancel. Since M , and therefore S , is non decreasing, we get

$$S_{n+2} \leq S_{n+1} \left(1 + \frac{A_r^{(1)}}{n+2} + \frac{A_r^{(0)}}{(n+2)(n+1)} \right) \leq S_{n+1} \left(1 + \frac{A_r^{(1)} + A_r^{(0)}}{n+2} \right).$$

Call $A_r^{(1)} + A_r^{(0)}$ a , then $S_{n+2} \leq S_1 \prod_{k=0}^n \left(1 + \frac{a}{k+2} \right)$, where this symbol is a product like how sigma is a sum. Although these inequalities assumed n was “large enough”, we just have to have multiplication by a constant to deal with the terms

that are not large enough, so it will not affect the radius of convergence argument. We can turn this product into a sum by taking logs on both sides to obtain

$$\ln(S_{n+2}) \leq \ln(S_1) + \sum_{k=0}^n \ln\left(1 + \frac{a}{k+2}\right)$$

Now $\ln(1+x) \leq x$ for all $x > -1$ because $\ln(1+x) - x$ has a derivative of $\frac{1}{1+x} - 1$ which is positive when x is between -1 and 0 and negative when x is > 0 . $\ln(1+x) - x$ has a stationary point at $(0, 0)$, and because it is increasing before that and decreasing after that, it is never positive. Therefore,

$$\ln(S_{n+2}) \leq \ln(S_1) + \sum_{k=0}^n \frac{a}{k+2} < \ln(S_1) + \sum_{k=0}^n \frac{a}{k+1} = \ln(S_1) + \sum_{k=1}^{n+1} \frac{a}{k}$$

H_n is commonly used as shorthand for the quantity $\sum_{r=1}^n \frac{1}{r}$. We have that $\ln(S_{n+2}) \leq \ln(S_1) + a(H_{n+1}) \leq \ln(S_1) + a(H_n) + a$

$H_n \leq 1 + \ln(n)$. Why? Suppose, for example, $n = 5$. Then $1 + \ln(n)$ is given by this blue area with the graph being $y = \frac{1}{x}$ (easily shown by integration)

Figure 4 shows an area equal to $1 + \ln(n)$ for $n = 5$ as an example using the area bounded by $y = 0, y = 1, y = \frac{1}{x}, x = 0$ and $x = 5$.

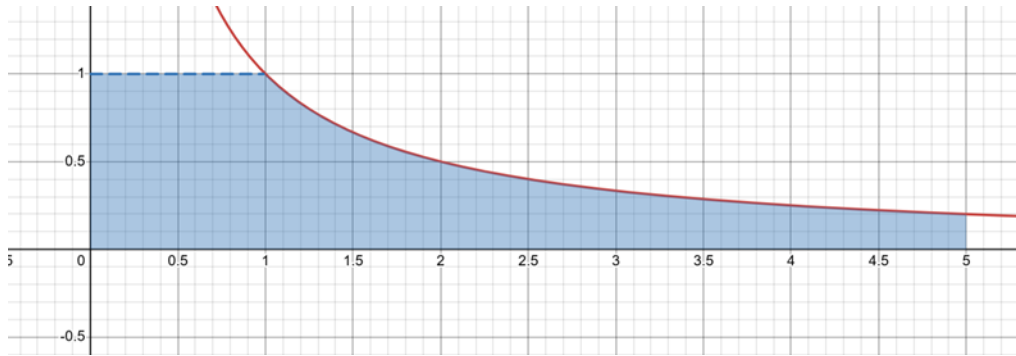


Figure 4

And H_n is this area which is clearly smaller:

Figure 5 shows an area equal to $H(n)$ for $n = 5$ as an example by using the area bounded by $x = 0, x = 5, y = 0, y = \frac{1}{\text{smallest integer} > x}$, visually demonstrating that $H(n) < 1 + \ln(n)$

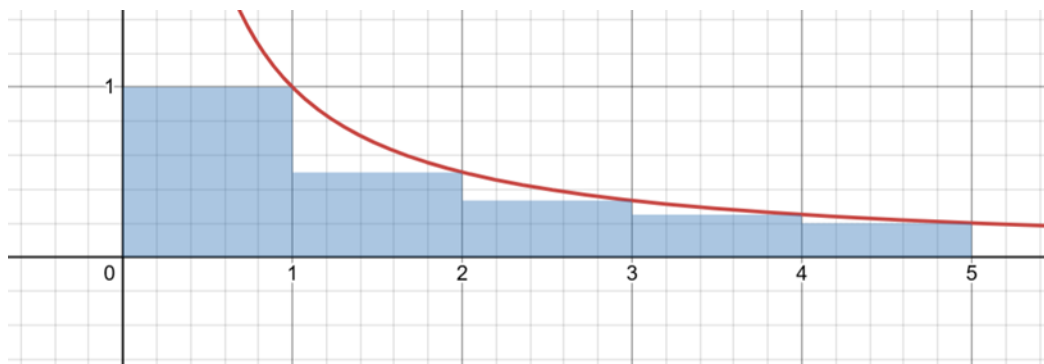


Figure 5

$$\ln(S_{n+2}) \leq c + a(H_n) \leq c + a(1 + \ln(n)) \leq c + a(\ln(n))$$

Where in the last step I have renamed c : What matters is it's still a constant. Un-logging both sides gives

$$S_{n+2} \leq cn^a$$

Where again, c is still some constant.

We now have that $S_n \leq c(n-2)^a \leq cn^a$

Why do all this? Because the radius of convergence of our power series is given by

$$\frac{1}{\limsup_{n \rightarrow \infty} |c_n|^{1/n}}.$$

Now using the definitions, this result, and the fact that $S_n \geq M_n \geq 0$, we have that

$$|c_n|^{1/n} \leq \left(\frac{M_n}{|x|^n} \right)^{\frac{1}{n}} \leq \left(\frac{S_n}{|x|^n} \right)^{\frac{1}{n}} \leq \left(\frac{cn^a}{|x|^n} \right)^{\frac{1}{n}} = \frac{c^{1/n} n^{a/n}}{|x|}$$

So, since $c^{1/n} n^{a/n} \rightarrow 1$ (You can take logs and use L'hôpital, proved in the video, to check this), the lim sup of this is no larger than $\frac{1}{|x|}$. Since x could be anything less than R (where R is the common radius of convergence that we've assumed $|x|$ is less than the entire time), the lim sup is no larger than $\frac{1}{R}$, so the reciprocal is no smaller than R , so done (at last). The series solution does in fact converge within the common radius of convergence of the coefficients of a 2^{nd} order linear differential equation.

□

2.5 Other results

Theorem. L'hôpital's rule works if we impose some conditions that the functions are reasonable (see the technical results document for a more detailed statement of the conditions and a fully rigorous proof)

Proof. I have a video on this which shows a visual argument for the $\frac{0}{0}$ case and then at the end shows images of a proof of the other cases from that case.

□

A tricky integral problem

Pick the standard continuous branch of the logarithm for $\operatorname{Re}(z) > 0$ and for $x > 0$ a real number let $F(x) = \frac{1}{2i} \ln \left(\frac{x-i}{x+i} \right)$. Direct computation gives $F'(x) = \frac{1}{1+x^2}$. Therefore $F(x) = \arctan(x) + c$ on that interval. But we also have $\int \frac{1}{x^2+1} dx = \frac{1}{2i} \ln \left(\frac{x-i}{x+i} \right) + c$, so we have two different answers. However, it turns out these are secretly equivalent, and we will see how this comes from the $+c$. A simpler example of this is how the integral of $\frac{1}{2x}$ gives both $\frac{\ln(x)}{2} + c$ and $\frac{\ln(2x)}{2} + c$ depending on the method used to solve it. The resolution to this second example is that the two answers actually differ by $\frac{\ln(2)}{2}$ by logarithm properties, which is a constant.

We can do the same for the first example: Since $\left| \frac{x-i}{x+i} \right| = \frac{x^2+1}{x^2+1} = 1$ if x is real,

$$\begin{aligned} \ln \left(\frac{x-i}{x+i} \right) &= i * \arg \left(\frac{x-i}{x+i} \right) = i * \arg \left(\frac{ix+1}{ix-1} \right) = i * \arg(ix+1) - i * \arg(ix-1) \\ &= i * \arctan(x) - i * \arctan(-x) = 2i \arctan(x) \end{aligned}$$

So $\frac{1}{2i} \ln \left(\frac{x-i}{x+i} \right) = \arctan(x)$, not even different by a constant.

So, two different answers could be secretly equivalent: This often happens when trig identities are involved. Note that if $|x| = 1$ then x can be written as $e^{i \arg(x)}$, and $\arg(ix+1) = \arctan(x)$ because this can be shown geometrically.

2.6 Arc length

Proposition. Arc length of a parametric curve does not depend on how you parametrize it, provided each point on the curve is only traversed once.

Proof. Let $x(t)$ and $y(t)$ parametrize a curve and have continuous derivatives from a to b and $x(f(t))$ and $y(f(t))$ parametrize the same curve and f is some increasing function from $[a, b]$ to $[a, b]$ with a continuous derivative. Suppose also that the parametrization has $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 > 0$ everywhere. Then by simple integration by substitution,

$$\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_a^b \sqrt{\left(\frac{dx}{df} \frac{df}{dt}\right)^2 + \left(\frac{dy}{df} \frac{df}{dt}\right)^2} dt = \int_a^b \sqrt{\left(\frac{dx}{df}\right)^2 + \left(\frac{dy}{df}\right)^2} \frac{df}{dt} dt = \int_a^b \sqrt{\left(\frac{dx}{df}\right)^2 + \left(\frac{dy}{df}\right)^2} df$$

It then remains to show that for any 2 parametrizations of the curve that are regular (ie $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 > 0$ everywhere), one can be written as a function of the other. Let x^* and y^* be a different parametrization

At each point, either $\frac{dx}{dt}$ or $\frac{dy}{dt}$ is non-zero and is hence non-zero in some interval by continuity, say $\frac{dx}{dt}$. Then x has a local continuously differentiable inverse x^{-1} so $x^{-1}(x^*(t))$ is a continuously differentiable function that works in some interval and is uniquely determined. Call this $u_x(t)$. We can define $u_y(t)$ similarly. It turns out these coincide since $(x, y)(t)$ and $(x^*, y^*)(t)$ are both one-to-one maps so it is clear that $(x, y)^{-1}(x^*, y^*)(t)$ is a one-to-one map u from reals to reals, and we just have to show that it is continuously differentiable.

On overlapping intervals where both u_x and u_y are defined, they must be strictly equal because the mapping to the curve is one-to-one. Therefore, these local maps seamlessly glue together to form a globally continuously differentiable function u , as the overlapping open intervals cover the entire domain, preventing any jumps or undefined derivatives at the patch boundaries.

□

3 Conic sections

3.1 Why they come from cones

They are called conic sections because you get them from slicing a cone, but typically at A level this is not proven. If a cone has equation $c_1 z^2 = x^2 + y^2$ and a plane has equation $x + c_2 z = 1$ (where we have rotated and scaled the figure so that the plane intersection with $z = 0$ closest to the origin has been standardized to the point $(1, 0)$ in the xy plane). Then we can make a substitution (specifically we rearrange the second equation for z and substitute that into the first equation): $A(1 - x)^2 = x^2 + y^2$ where A is some constant. If $A = 1$ this cancels to give $y^2 = 1 - 2x$ which is a parabola. This means a parabola is what you would see if you were to look at the plane from the z axis (ie, looking down from a point like $(0, 0, 10000)$). If you were to look at the plane from its perpendicular axis, it would stretch out, below is an illustration of what I mean, we see that if we ignore perspective issues and act like we are simply seeing a projection, the plane looks like a contracted version of itself when viewed from an axis not perpendicular to it.

Figure 6 shows a plane viewed from above and from the side, with its gridlines, so you can see my point.

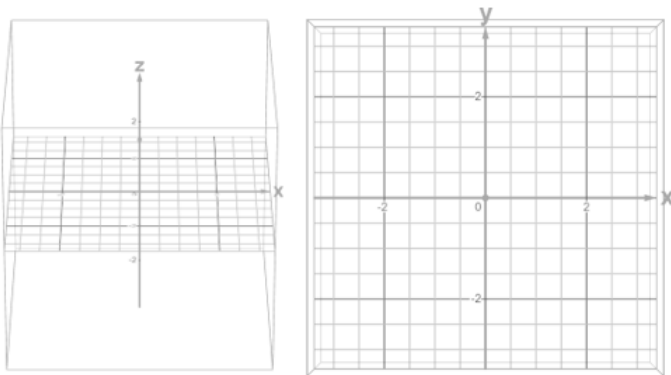


Figure 6

Stretching a parabola still gives a parabola, if the x or y is scaled it's still a parabola.

Similarly, if $A < 1$, we get an ellipse equation, if $A > 1$ we get a hyperbola equation. In both cases the stretching argument still applies. Therefore we know that all three conic sections are gotten by slicing a cone.

Also, you can consider a parabola to be like a “limit” of an ellipse as it gets more and more stretched out, by slowly tilting the plane intersecting the cone so A is 0.99, 0.999, 0.9999, etc

We can also easily verify, using trig identities, that the parametric equations for an ellipse, parabola, or hyperbola satisfy the cartesian equations for them. The only thing we need to be careful of is that if we use $(A \cosh(t), B \sinh(t))$ as the parametric equation for a hyperbola, then $\cosh$ can only take positive values if t is real, so we should either add a second branch with $-A \cosh(t)$, allow complex t , or use $\tan$ and $\sec$.

By the way don't consider planes coming from the apex of the cone those are degenerate.

3.2 Focus distance properties

Now we will prove that in an ellipse and hyperbola, the sum and difference of the distances of any point on the curve from the foci respectively is constant, this is the defining feature of the foci. For an ellipse we start by assuming the sum of the distances from the foci at $(-c, 0)$ and $(c, 0)$ is constant and equal to $2a$ with $a > c$ then we prove that the equation we get is the equation for an ellipse.

$$\begin{aligned}\sqrt{(x+c)^2 + y^2} + \sqrt{(x-c)^2 + y^2} &= 2a \\ \sqrt{(x+c)^2 + y^2} &= 2a - \sqrt{(x-c)^2 + y^2} \\ (x+c)^2 + y^2 &= \left(2a - \sqrt{(x-c)^2 + y^2}\right)^2 \\ x^2 + 2cx + c^2 + y^2 &= 4a^2 - 4a\sqrt{x^2 - 2cx + c^2 + y^2} + x^2 - 2cx + c^2 + y^2 \\ 4cx &= 4a^2 - 4a\sqrt{x^2 - 2cx + c^2 + y^2} \\ a\sqrt{x^2 - 2cx + c^2 + y^2} &= a^2 - cx \\ a^2(x^2 - 2cx + c^2 + y^2) &= a^4 - 2a^2cx + c^2x^2 \\ a^2x^2 + a^2c^2 + a^2y^2 &= a^4 + c^2x^2 \\ x^2(a^2 - c^2) + a^2y^2 &= a^2(a^2 - c^2) \\ \frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} &= 1\end{aligned}$$

Note: the steps where we square both sides can be reversed since we always assume both sides to be positive - they are always sums of squares of stuff or square roots of such expressions. Therefore we have equivalence between the equation for the ellipse in “pythagorean” form and focus distance form.

In a hyperbola we do nearly the identical work. You should be able to figure it out by following the same steps as in the ellipse proof, but here it is anyway.

Note that the equation we are about to write which is $\sqrt{(x+c)^2 + y^2} - \sqrt{(x-c)^2 + y^2} = 2a$ is something which we only claim gives the right branch of the hyperbola. For the left branch we would actually want $\sqrt{(x-c)^2 + y^2} - \sqrt{(x+c)^2 + y^2} = 2a$. In the algebra below we actually must have $x \geq 0$ for the first line to have a positive right hand side so we restrict to $x \geq 0$ and would restrict to $x \leq 0$ for the other branch. The proof would then proceed with signs reverse until the sign-reversed terms turn out to cancel as they do in the last step of the algebra below:

$$\begin{aligned}\sqrt{(x+c)^2 + y^2} - \sqrt{(x-c)^2 + y^2} &= 2a \\ \sqrt{(x+c)^2 + y^2} &= 2a + \sqrt{(x-c)^2 + y^2} \\ (x+c)^2 + y^2 &= \left(2a + \sqrt{(x-c)^2 + y^2}\right)^2 \\ x^2 + 2cx + c^2 + y^2 &= 4a^2 + 4a\sqrt{x^2 - 2cx + c^2 + y^2} + x^2 - 2cx + c^2 + y^2\end{aligned}$$

$$\begin{aligned}
4cx &= 4a^2 + 4a\sqrt{x^2 - 2cx + c^2 + y^2} \\
a\sqrt{x^2 - 2cx + c^2 + y^2} &= cx - a^2 \\
a^2(x^2 - 2cx + c^2 + y^2) &= a^4 - 2a^2cx + c^2x^2 \\
a^2(x^2 + c^2 + y^2) &= a^4 + c^2x^2
\end{aligned}$$

From here, the steps are the same as before, except now the equation we get at the end is a hyperbola equation because $c > a$.

In a parabola $y^2 = 4ax$ the distance from the focus $(a, 0)$ to a point on the parabola is given by $\sqrt{(x - a)^2 + y^2}$, and $x = \frac{y^2}{4a}$ so we have $\sqrt{(\frac{y^2}{4a} - a)^2 + y^2}$. We can simplify this to $\sqrt{\frac{y^4}{16a^2} - \frac{y^2}{2} + a^2 + y^2}$ which is $\sqrt{\frac{y^4}{16a^2} + \frac{y^2}{2} + a^2}$ which is $a + \frac{y^2}{4a}$ which is $a + x$ which is the distance from the relevant point on the parabola to the directrix.

3.3 Eccentricity properties

Now, we prove properties of eccentricity. Suppose that the eccentricity of an ellipse or hyperbola is the value of e such that $x^2 + \frac{y^2}{1-e^2} = 1$, then we will prove that e is both the ratio between the distance between the foci and the length of the long diameter of the ellipse or the distance between the parts of the hyperbola, and the ratio of the distance of a focus and the corresponding directrix. For the first one, note that if we consider e to be $\frac{c}{a}$ in the proofs above the final equations can be simplified to be $x^2 + \frac{y^2}{1-e^2} = 1$ and also that by picking the point on the curve in question to be a point on the x axis it becomes clear that $2a$, our constant distance is indeed the length of the long diameter of the ellipse or the distance between the parts of the hyperbola.

For the other property, if we have a point on the curve, we let d_1 be the distance from that point to the focus and d_2 be the distance from that point to the corresponding directrix. Note also that even if the denominator of x is not 1 both properties hold by scaling. We have that $d_1 = \sqrt{(x - e)^2 + y^2}$ by pythagoras and that $d_2 = \frac{|1 - ex|}{e}$. Substituting in $y^2 = (1 - x^2)(1 - e^2)$ which comes from rearranging the equation for the curve we get that

$$d_1 = \sqrt{(x - e)^2 + (1 - x^2)(1 - e^2)} = \sqrt{x^2 - 2xe + e^2 + 1 - x^2 - e^2 + x^2e^2} = \sqrt{-2xe + 1 + x^2e^2} = xe - 1 = ed_2$$

as required.

4 Complex roots of real polynomials

Proposition. Let $f(x)$ be a polynomial with real coefficients and suppose that z is a root, then the conjugate of z is also a root.

Proof. Note that $f(x)$ is a sum of constants times x^n . So let's say $f(x) = \sum_{r=0}^n a_r x^r$. Then let $x = re^{i\theta}$, so we have $f(x) = \sum_{t=0}^n a_t r^t e^{it\theta}$. Since the a_n coefficients are real, the complex conjugate of the right hand side is

$$\sum_{t=0}^n a_t r^t e^{-it\theta} = \sum_{t=0}^n a_t (\bar{x})^t = f(\bar{x})$$

Therefore, for any x , we have

$$f(\bar{x}) = \overline{f(x)}$$

Therefore,

$$f(\bar{z}) = \overline{f(z)} = \bar{0} = 0$$

□

5 Recurrence relations

We will not really do any work here because the parts of this that are used as a black box in some specifications can be proven using the same ideas as the differential equations from earlier. Perhaps this is unsurprising since the methods are very similar for solving both types of problems. If you use the analogous substitution for the second order case, then you will get the right complementary function, and the same result that says the thing about the general solution being complementary function plus particular solution also applies and can be proven the same way. It remains to verify the results in the tables of particular solutions given in some textbooks. However, it is easy to verify that you will get a solvable equation to solve for the constants as a result of plugging them in.

I figure I should do one from first principles/properly just to demonstrate. I admit that it is a bit longer but at least it doesn't feel like utter magic that you have to memorize and trust. Let's solve

$$u_n - 7u_{n-1} + 12u_{n-2} = 3^n$$

Let's use the substitution $t_n = u_n - 3u_{n-1}$ since 3 is one of the roots of the characteristic equation. We get

$$t_n - 4t_{n-1} = 3^n$$

Let's also say $w_n = \frac{t_n}{4^n}$. This gives

$$\begin{aligned} w_n 4^n - 4 * 4^{n-1} w_{n-1} &= 3^n \\ w_n - w_{n-1} &= \left(\frac{3}{4}\right)^n \end{aligned}$$

This gives us, by geometric series formula,

$$w_n = w_0 + \sum_{r=1}^n \left(\frac{3}{4}\right)^r = w_0 + 3 \left(1 - \left(\frac{3}{4}\right)^n\right)$$

Thus

$$t_n = 4^n \left(c + 3 \left(1 - \left(\frac{3}{4} \right)^n \right) \right)$$

where c is a constant. Thus we want to solve

$$t_n = u_n - 3u_{n-1} = 4^n \left(c + 3 \left(1 - \left(\frac{3}{4} \right)^n \right) \right)$$

This may look scary but it isn't. Let's expand it out

$$u_n - 3u_{n-1} = c * 4^n + 3 * 4^n - 3 \left(\frac{3}{4} \right)^n 4^n$$

Note that $c * 4^n + 3 * 4^n$ is still $c * 4^n$ where c is some constant, and $3 \left(\frac{3}{4} \right)^n 4^n = 3 * 3^n$. Thus we simply have

$$u_n - 3u_{n-1} = c * 4^n - 3 * 3^n$$

Let's substitute (to get it into a $d_n - d_{n-1}$ form) that $d_n = \frac{u_n}{3^n}$, so that $u_n = 3^n d_n$. This gives

$$\begin{aligned} 3^n d_n - 3^n d_{n-1} &= c * 4^n - 3 * 3^n \\ d_n - d_{n-1} &= c * \left(\frac{4}{3} \right)^n - 3 \end{aligned}$$

Note that if we are summing $-3 * 1^n$ then what happens is different from if we are summing $-3 * a^n$ for $a \neq 1$, and THIS, is not hard to see why. This is all that is going on, not the “because I said so” I was joking about in level 5, even though THAT IS LITERALLY HOW MANY TEXTBOOKS TEACH IT JUST MORE NICELY.

Ok anyway we have

$$\begin{aligned} d_n &= d_0 + \sum_{r=1}^n \left(c * \left(\frac{4}{3} \right)^r - 3 \right) \\ &= d_0 - 3n + c * \sum_{r=1}^n \left(\frac{4}{3} \right)^r \\ &= d_0 - 3n + 4c * \left(\left(\frac{4}{3} \right)^n - 1 \right) \end{aligned}$$

again by geometric series formula

$$= A - 3n + B \left(\frac{4}{3} \right)^n$$

where A and B are arbitrary constants.

Reversing the substitution, $u_n = 3^n d_n$ so the solution to our original recurrence relation is

$$u_n = A * 3^n + B * 4^n - 3n * 3^n$$

Notice how everything came out by logic instead of magic.